

# Mohamed Nisfan

Science and Technology undergraduate specializing in Mechatronics

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📍 33/A, Meepitiya, Nawalapitiya 🔗 Portfolio 🌐 Mohamed Nisfan



## Profile

Science and Technology undergraduate specializing in Mechatronics with strong experience in full-stack web development, backend systems, and machine learning applications. Skilled in building scalable web applications using React.js, Next.js Node.js MongoDB, and Firebase. Experienced in developing e-commerce platforms, IoT systems, and AI-driven applications. Passionate about developing intelligent, data-driven systems, and creating efficient data-driven solutions.

## PROFESSIONAL EXPERIENCE

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| <b>Undergraduate Trainee</b><br><i>C.W. Mackie PLC - Scan Division</i><br>Designed mechanical components and assemblies using SolidWorks, prepared AutoCAD drawings, and maintained BOMs and technical documentation while supporting equipment performance improvements. | 07/2025 – 10/2025<br>Horana, Sri Lanka |
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## EDUCATION

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| <b>Bsc. Hons. in Science and Technology (Mechatronics specialization)</b><br><i>Uva Wellassa University</i>      | 09/2021 – 08/2025 |
| <b>Diploma in Management</b><br><i>Uva Wellassa University</i>                                                   | 05/2022 – 05/2023 |
| <b>Certificate in Human Resource Management</b><br><i>Chartered Institute of Personnel Management, Sri Lanka</i> | 06/2020 – 12/2020 |

## SKILLS & EXPERTISE

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| <b>Programming Languages</b><br>JavaScript, TypeScript, Python, C / C++, HTML & CSS                                                                                                                   | <b>Backend Development</b><br>Node.js, Express.js, Authentication Systems, REST API Development                                  |
| <b>Frontend Development</b><br>React.js, Next.js, TailwindCSS                                                                                                                                         | <b>Deployment &amp; Tools</b><br>Git & GitHub, Vercel, Supabase, Cloudinary, Arduino IDE                                         |
| <b>Machine Learning and Data Science</b><br>Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib, Deep Boltzmann Machine (DBM), Convolutional Neural Networks (CNN) | <b>Embedded Systems &amp; IoT</b><br>ESP32 programming, Sensor integration, Real-time machine monitoring, IoT data communication |
| <b>Soft Skills</b><br>Communication, Teamwork, Problem-solving, Analytical Thinking                                                                                                                   |                                                                                                                                  |

## Languages

• Sinhala • English • Tamil

## EXTRACURRICULAR ACTIVITIES

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### Director of Creation Team

2024 – 2025

Career Guidance Club of Uva Wellassa University

### Public Visibility Head

2023 – 2024

IEEE Computer Society Uva Wellassa University

Student Branch Chapter

## PROJECTS

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### PC Corner - Full Stack E-Commerce Platform

*Developed a full-stack e-commerce website for a computer hardware store with features including product catalog management, shopping cart, order processing, and customer authentication. Implemented backend APIs for product management, order handling, and user authentication using Node.js and MongoDB. Integrated Cloudinary for image storage and deployed the platform online.*

Node.js | Express.js | MongoDB | React | Cloudinary | REST APIs | Vercel

### Restaurant POS & Billing System

*Developed a web-based POS system for restaurants featuring real-time billing, order management, product tracking, and daily sales reporting. Implemented offline order processing with automatic synchronization when the internet connection is restored. Integrated thermal printer communication using WebUSB API for direct receipt printing.*

React | TypeScript | Supabase | PostgreSQL | Vite | WebUSB API | Vercel

### Library Management CRUD App using Next.JS

*Developed a full-stack Library Management System using Next.js to perform efficient CRUD operations for managing book records, with a focus on modular UI design and dynamic data handling.*

Next.js | React | Tailwind CSS | Node.js | MongoDB | Mongoose

### Online Printer Management System using Python

*Developed an intuitive web-based printer management system using Django, Python, HTML, and CSS, featuring secure authentication, file uploads, customizable printing options, dynamic cost calculation, and real-time print job monitoring.*

Python | HTML | CSS | Javascript

### Real-Time Machine Health Monitoring Using Deep Boltzmann Machine

*Developed an intelligent monitoring system to analyze machine health using real-time sensor data including voltage, current, temperature, and vibration. Implemented a Deep Boltzmann Machine (DBM) model to detect anomalies and predict potential machine failures. The system collects sensor data using microcontrollers and processes it using Python for fault detection.*

Embedded | Python | Arduino IDE | Deep Learning (DBM) | Sensors | Firebase

### Machinery Identification using CNN (OpenCV)

*Developed a machine learning-based image classification model to identify workshop machinery using CNNs. Trained with augmented image datasets and deployed for real-time predictions using OpenCV and Keras.*

Python | TensorFlow | Keras | OpenCV | CNN | Image Classification | Machine Learning

### Sign Language Detection using Python & Scikit-learn

*Developed a machine learning-based application to recognize and interpret sign language gestures using Python and Scikit-learn, enabling real-time gesture classification for improved communication accessibility.*

Python | OpenCV | MediaPipe | Scikit-learn | Numpy

## References

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**DR. THUSHARA EKANAYAKE**, Senior Lecturer,

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**Mr. DHANUSHKA ARANDARA**, Group Head of Engineering, C.W.Mackie Group

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